Rohanth Gundu

AI/ML Engineer

rohanthjobs@gmail.com | +1 (314) 642-0677 | San Francisco, CA | LinkedIn

SUMMARY

Results-driven AI/ML Engineer with 6+ years of experience building, deploying, and optimizing production-ready machine learning systems. Expertise in Python, deep learning, NLP, computer vision, MLOps, and cloud infrastructure (AWS, GCP, Azure). Skilled at designing end-to-end ML pipelines, implementing model deployment and monitoring solutions, and translating complex AI workflows into measurable business outcomes. Adept at collaborating with cross-functional teams to deliver scalable, real-time AI solutions.

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Perplexity **01/2024 – Present | USA**

- Architected retrieval-augmented generation (RAG) pipelines processing 1M+ monthly queries, integrating semantic search and LLM-driven contextual retrieval to boost accuracy and responsiveness by 40%.
- Developed real-time data ingestion and feature extraction pipelines using Kafka, gRPC, and Spark to support continuous model training and inference under sub-60 ms latency SLAs.
- Built and fine-tuned custom embedding models with PyTorch and TensorFlow, storing high-dimensional vectors in Pinecone for scalable retrieval and personalized recommendations.
- Designed GPU-optimized inference microservices on Kubernetes (AWS, Terraform, Helm), automating scaling for large transformer-based models in production.
- Implemented event-driven ML workflows via NATS JetStream and Redis to orchestrate asynchronous model predictions and ensure reliable, low-latency AI responses.
- Integrated multimodal AI pipelines combining NLP, speech recognition, and sentiment analysis using GStreamer, FFmpeg, and deep learning diarization models for real-time audio analytics.
- Developed RESTful inference APIs using FastAPI and Flask to serve LLM, classification, and recommendation models with robust error handling and high throughput.
- Monitored and optimized ML models in production with Prometheus, Grafana, and MLflow to detect drift, maintain stability, and guide retraining cycles.
- Collaborated with data scientists and product teams to enhance user-facing dashboards and deliver AI-driven insights such as investor alerts and contextual recommendations.
- Researched and implemented improvements in model architecture, embeddings, and inference pipelines, contributing to company-wide AI performance benchmarks.

AI/ML Engineer, Meta **06/2019 – 10/2023 | India**

- Designed and deployed large-scale ML models for recommendation systems, NLP, and computer vision pipelines, improving user engagement by 18%.
- Built semantic retrieval systems using Hugging Face Transformers, LangChain, and FAISS vector databases, reducing search latency by 45%.
- Led end-to-end LLM-based question-answering pipelines, integrating prompt engineering, retrieval augmentation, and deployment for internal knowledge access.
- Developed MLOps pipelines using Airflow, MLflow, and Kubernetes (EKS), integrating feature engineering, hyperparameter tuning, and model monitoring.
- Containerized ML workflows with Docker and deployed on cloud infrastructure (AWS, GCP), enabling horizontal auto-scaling and high availability.
- Engineered ETL pipelines and feature stores for structured and unstructured datasets, ensuring high-quality input for ML models.
- Implemented model monitoring and explainability using SHAP, LIME, and drift detection to ensure responsible AI practices.
- Deployed RESTful ML services for production, handling millions of inference requests per day with low latency.
- Collaborated in cross-functional agile squads with product, research, and engineering teams to optimize ML solutions for scalability and business impact.
- Mentored junior engineers on AI/ML modeling, MLOps best practices, and responsible AI guidelines, accelerating team onboarding and productivity.

EDUCATION

Master's in Computer Science
Saint Louis University, Missouri, USA

SKILLS

Languages & Frameworks: Python (FastAPI, Flask), SQL, R, Java, PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, spaCy, Hugging Face Transformers, LangChain, NumPy, Pandas, Matplotlib, Seaborn.
Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning (CNNs, RNNs, Transformers), Large Language Models (LLMs), NLP (NER, Summarization, Sentiment Analysis), Computer Vision, Generative AI, RAG, Recommendation Systems, Anomaly Detection, Feature Engineering, Hyperparameter Tuning, Model Evaluation.
MLOps & DevOps: MLflow, Airflow, DVC, CI/CD Pipelines, Docker, Kubernetes (EKS, OpenShift), Helm, ArgoCD, Git, GitOps, Terraform, Ansible, Cloud Functions, Experiment Tracking, Model Registry, Canary Releases.
Cloud Platforms & Big Data: AWS (S3, EC2, Lambda, SageMaker), GCP (Vertex AI, BigQuery, Dataflow), Azure ML, Snowflake, Delta Lake, Feature Stores, Data Lakes, Spark, Kafka, Event-Driven Architectures, REST APIs.
Monitoring, Explainability & Responsible AI: SHAP, LIME, Captum, Model Drift Detection, Bias/Fairness Audits, Performance Dashboards, Responsible AI Practices, Model Transparency.
Data Engineering & Streaming: ETL/ELT Pipelines, Real-time Data Ingestion, gRPC, WebSockets, NATS JetStream, Redis, Delta Lake, Apache Spark, Spark Streaming, Flink.
Visualization & Reporting: Looker, Tableau, Power BI, Jupyter Notebooks, Matplotlib, Seaborn, Plotly, Interactive Dashboards for ML Metrics.
Workflow & Methodologies: Agile, Scrum, Cross-functional Collaboration, Design Thinking, AI Governance, Data Privacy Compliance (GDPR, CCPA), A/B Testing, Prompt Engineering.